1. Blume-Capel Model: Consider this extension of the Ising model on a d-dimensional lattice with z nearest neighbours.

$$H = -J \sum_{\langle i,j \rangle} S_i S_j + \Delta \sum_{i=1}^{N} S_i^2 - h \sum_{i=1}^{N} S_i$$

with $S_i = \{-1, 0, +1\}$. Here $\sum_{\langle i,j \rangle}$ denote a sum over nearest neighbours only, as usual.

   (a) Solve the for the partition function in the mean field approximation.
   Hint: Do not approximate the $(\Delta \sum_{i=1}^{N} S_i^2)$ term as this can be evaluated exactly. The free energy in mean field theory is

$$A = \frac{NJzm^2}{2} - k_B T N \ln\{1 + e^{-\beta\Delta}[2\cosh(\beta[Jzm + h])]\},$$

   where $m = \langle S_i \rangle$

   (b) Now expand the logarithm and write the free energy in Landau form with $h = 0$. What values of (T, κ), where $\kappa = e^{-\beta\Delta}$ predict (a) normal critical behaviour, (b) tricritical behaviour, (c) a first-order transition? Find the value of the transition temperature in each case.

2. Determine the critical exponent λ for the following functions $\epsilon \to 0$. Here $\epsilon = (T - T_c)/T_c$.

$$f(\epsilon) = A\epsilon^{1/2} + B\epsilon^{1/4}14 + C\epsilon$$
$$f(\epsilon) = A\epsilon^{-2/3}(\epsilon + B)^{2/3}$$
$$f(\epsilon) = A(\epsilon^2 + B^2)^{1/2}(\ln|\epsilon|)^2$$

   [Hint: If $f(\epsilon) \sim \epsilon^\lambda$, then $\lambda = \lim_{\epsilon \to 0} \frac{\ln f(\epsilon)}{\ln \epsilon}$.]

3. Problem 5.3, pg 165 Goldenfeld. [Click here to see the problem]
Hint for parts (c) and (d): Note that the system has three phases, high temperature phase ($M = 0$) and two low temperature phases ($M \neq 0$). Assume that the $M_{q_n} \neq 0$, $q_n \neq 0$ occurs only for $t < 0$ and treat the problem of minimization of M and q_n separately. If you are getting stuck, you can look up the paper referred to in the footnote but please follow Goldenfeld's notation and conventions.

4. First-order phase transitions: We discussed these in class but did not work out the algebra.

   (a) Consider the Landau free energy with a cubic invariant

$$\pounds = \frac{1}{2}\bar{a}_2(T - T_c)m^2 + \frac{1}{3}a_3 m^3 + \frac{1}{4}a_4 m^4$$

      i. Draw the free energy at different temperatures and convince yourself that there is indeed an abrupt transition
      ii. Determine the transition temperature
      iii. Determine the temperature window where the free energy has two coexisting minima

   (b) Now consider the Landau free energy with a sixth-order term

$$\pounds = \frac{1}{2}\bar{a}_2(T - T_c)m^2 + \frac{1}{4}a_4 m^4 + \frac{1}{6}a_6 m^6, \; a_4 < 0$$

      i. Plot the free energy as a function of temperature and show that one has a symmetry-breaking abrupt (first-order) transition.
      ii. Determine the transition temperature
      iii. Determine the temperature window where the free energy has two coexisting minima
      iv. Tricritical point: Show that we recover a continuous transition when $a_4 = 0$ and determine the critical exponents β, γ, and δ when $a_4 = 0$

   (c) Explain how and why you can get hysteresis? Can second-order transitions show hysteresis?

   (d) For both the cases, try to understand the location of m corresponding to the various minima and maxima of te Landau free energy for different temperatures, as discussed in class [see Arovas' notes].

Problem 4 need not be submitted.

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